Lecture 3
Series and Parallel Resistor Circuits

Slide Courtesy:
1.Electronics Technology Fundamentals, 3rd ed. Paynter and Boydell
2. Electronics Fundamentals 9th edition, Floyd/Buchla/Snyder

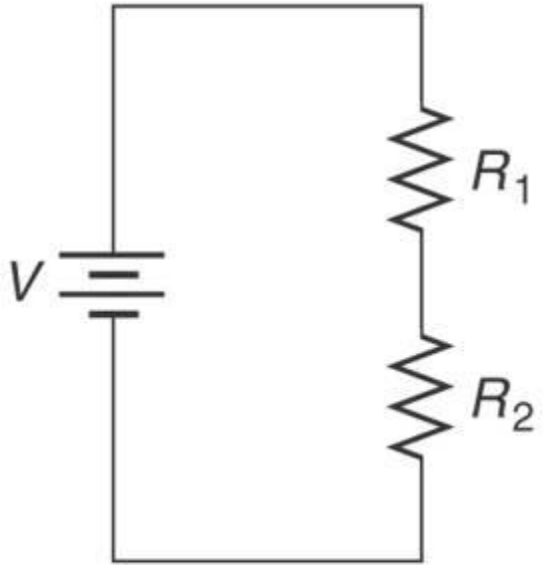
Outline

- **Series Resistor Circuit**
- Kirchhoff's Voltage Law (KVL)
- Voltage Divider

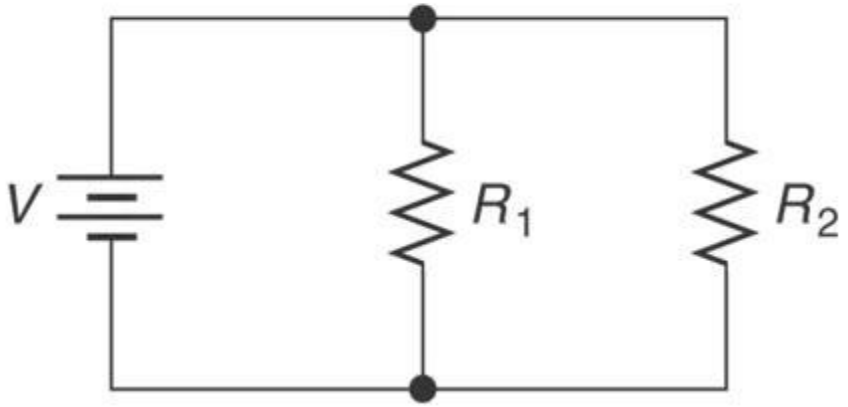
- **Parallel Resistor Circuit**
- Kirchhoff's Current Law (KCL)
- Current Divider

Series and Parallel Circuit

- **Most circuits, no matter how complex, contain components in series and/or in parallel.**



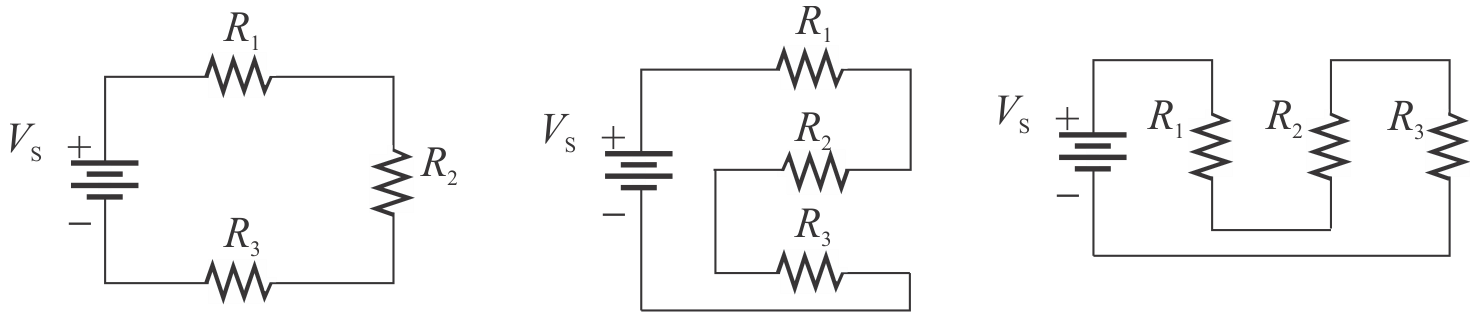
(a) Two-resistor series circuit



(b) Two-resistor parallel circuit

Series Circuit Characteristics

A ***series circuit*** is one that has only one current path.



Series Circuit Characteristics

- Total Series Resistance

$$R_T = R_1 + R_2 + \ldots + R_n$$

where

R_T = the total circuit resistance

R_n = the highest-numbered resistor in the circuit

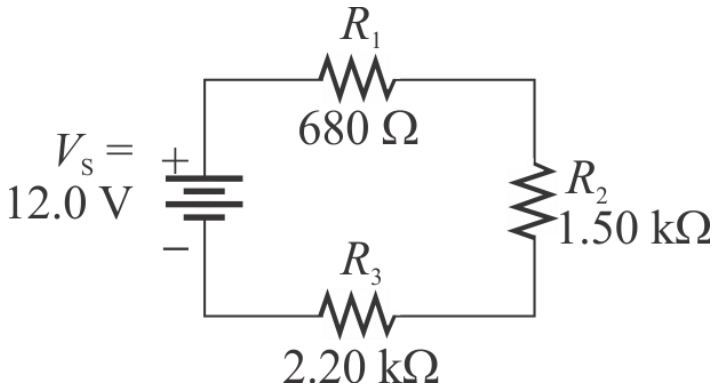
Series Circuit Example

The total resistance of resistors in series is

the sum of the individual resistors.

Example:

The resistors in a series circuit are 680 Ω, 1.50 kΩ, and 2.20 kΩ. What is the total resistance. R_T? 4.38 kΩ



Series Circuit Characteristics

- **Current Characteristics –** the current at any point in a series circuit must equal the current at every other point in the circuit
- Because there is only one path, the current everywhere is the same

- **Voltage Characteristics**

$$V_S = V_1 + V_2 + \ldots + V_n$$

- **Power Characteristics**

$$P_S = P_1 + P_2 + \ldots + P_n$$
$$= V_S I_T$$

Where

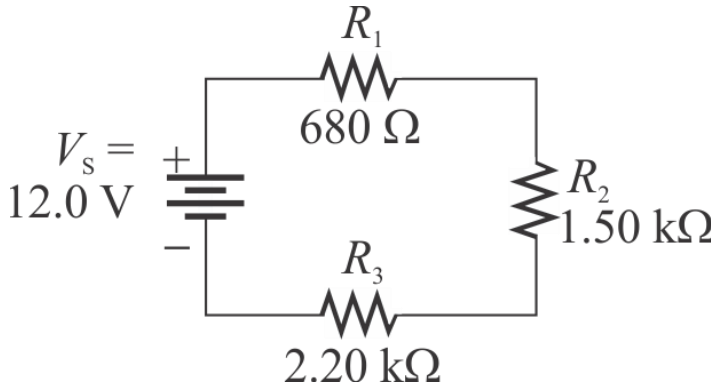
V_s = the source (or total) voltage

V_n = Voltage across the nth resistor in the circuit

P_S = the total power dissipated

Pn = the power that is dissipated across the highest numbered resistor in the circuit

Tabulating current, resistance, voltage and power is a useful way to summarize parameters in a series circuit.



Continuing with the previous example, complete the parameters listed in the Table.

I_1= 2.74 mA	R_1= 0.68 kΩ	V_1= 1.86 V	P_1= 5.10 mW
I_2= 2.74 mA	R_2= 1.50 kΩ	V_2= 4.11 V	P_2= 11.3 mW
I_3= 2.74 mA	R_3= 2.20 kΩ	V_3= 6.03 V	P_3= 16.5 mW
I_T= 2.74 mA	R_T= 4.38 kΩ	V_S= 12.0 V	P_T= 32.9 mW

Kirchhoff's Voltage Law (KVL)

Kirchhoff's voltage law (KVL) is generally stated as:

The algebraic sum of the voltages around any closed path is zero.

A mathematical shorthand way of writing KVL is $\sum_{i=1}^{n} V_i = 0$

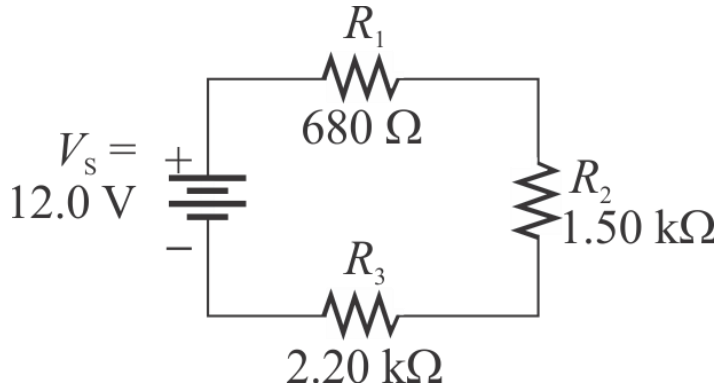
- The sum of the component voltages in a series circuit must equal the source voltage

$$V_S = V_1 + V_2 \ldots + V_n$$

- The following equation takes polarity into account

$$V_S + V_1 + V_2 + \ldots + V_n = 0 \text{ V}$$

Notice in the series example given earlier that the sum of the resistor voltages is equal to the source voltage.



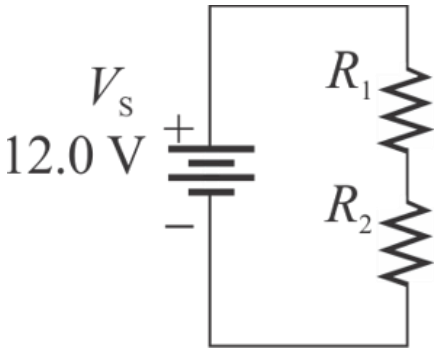
I_1= 2.74 mA	R_1= 0.68 kΩ	V_1= 1.86 V	P_1= 5.10 mW
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Voltage Divider Circuit

- The Voltage Divider Relationship
  - Voltage Divider – often used to describe a series circuit

Voltage Divider Rule:

The voltage drop across any given resistor in a series circuit is equal to the ratio of that resistor to the total resistance, multiplied by source voltage.



$$\frac{V_n}{V_S} = \frac{R_n}{R_T} \Rightarrow V_n = V_S \frac{R_n}{R_T}$$

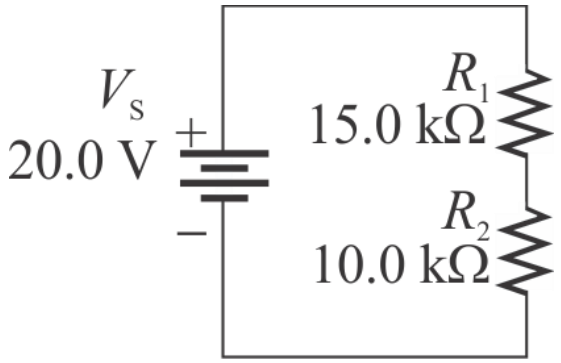
where

R_n = the resistor of interest

V_n = the voltage drop across R_n

(where n is the component number)

What is the voltage across R_2?



Solution:

The total resistance is 25 kΩ.
Applying the voltage divider formula:

$$V_2 = \left(\frac{R_2}{R_T}\right) V_S = \left(\frac{10\ \text{k}\Omega}{25\ \text{k}\Omega}\right) 20\ \text{V} = \quad 8.0\ \text{V}$$

Notice that 40% of the source voltage is across R2, which represents 40% of the total resistance.

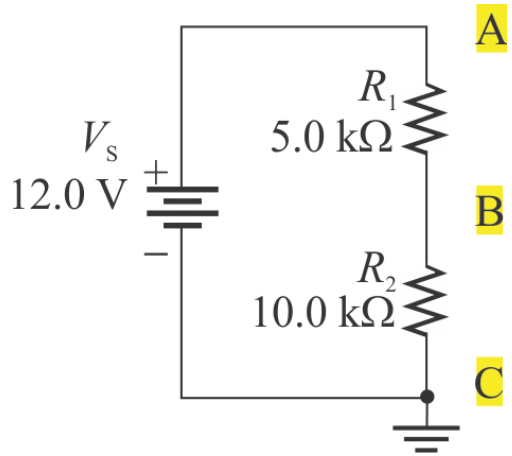
What is the voltage across R_1?

Apply KVL or Voltage divider rule:

KVL: $V_1 = Vs - V_2$

Voltage Measurements

- Voltage is relative and is measured with respect to another point in the circuit.
- Voltages that are given with respect to ground are shown with a single subscript.
- For example, $\boldsymbol{V_A}$ means the voltage at point A with respect to ground (called *reference ground*).
- $\boldsymbol{V_B}$ means the voltage at point B with respect to ground.
- $\boldsymbol{V_{AB}}$ means the voltage at point A relative to point B



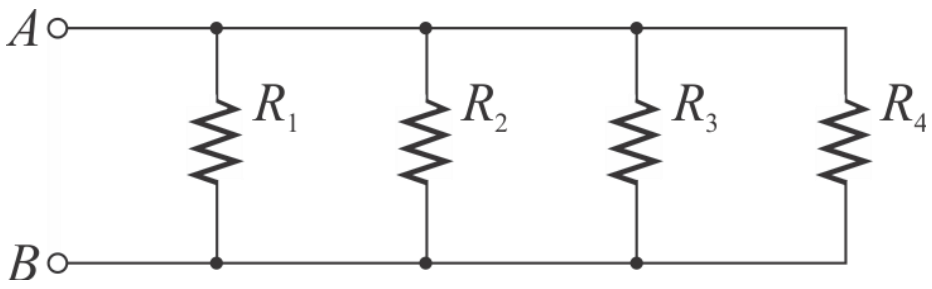
Question: What are V_A, V_B, and V_{AB} for the circuit shown?

$V_A = 12$ V $V_B = 8.0$ V $V_{AB} = +4.0$ V

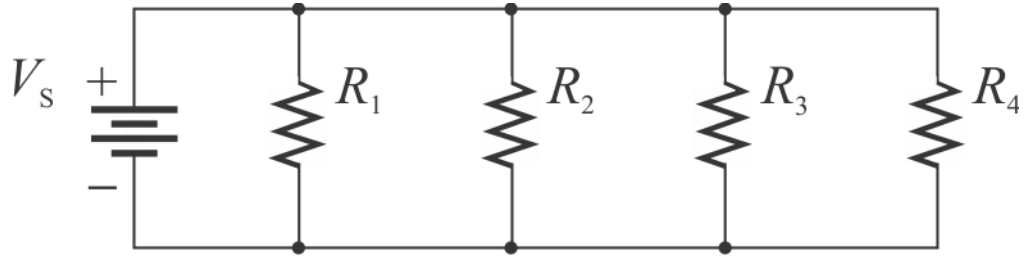
Follow-up: What is V_{BA} for the circuit shown? $V_{BA} = -4.0$ V

Resistors in Parallel

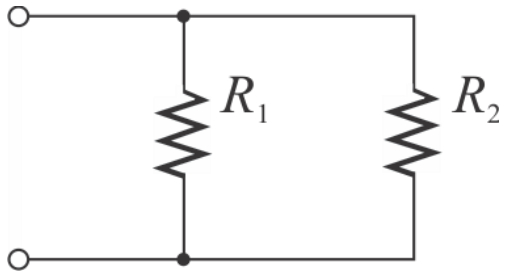
Resistors that are connected between the same two points are said to be in **parallel**.



A ***parallel circuit*** is identified by the fact that it has more than one current path (branch) connected to a common voltage source.



Two Parallel Resistors



The special case of resistance of two parallel resistors can be found by:

$$R_T = \frac{1}{\frac{1}{R_1} + \frac{1}{R_2}}$$

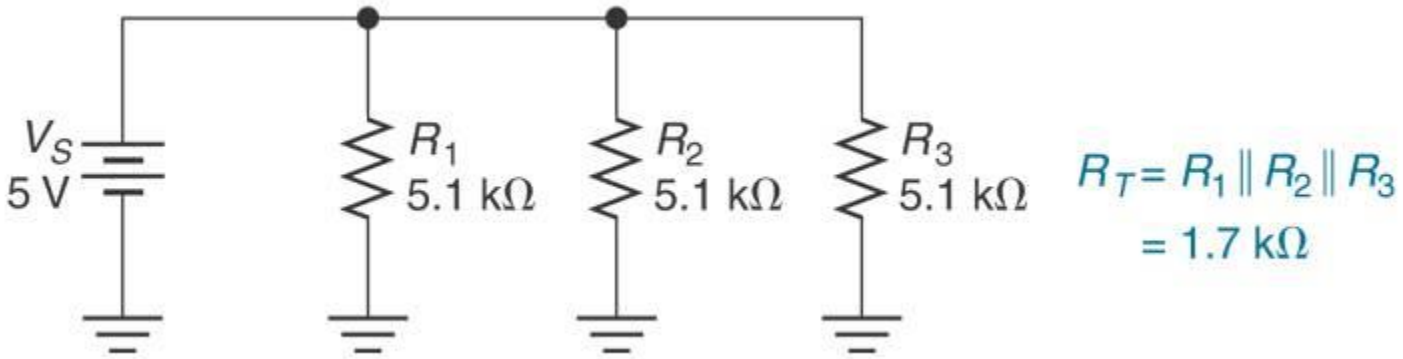
which can be reduced to

$$R_T = \frac{R_1 R_2}{R_1 + R_2}$$

This is called the product-over-sum rule.

Calculate the total resistance if R_1 = 27 kΩ and R_2 = 56 kΩ 18.2 kΩ

- Calculating Total Resistance When Branch Resistances are Equal – Divide the value of one resistor by the number of branches



$$R_T = \frac{5.1\text{k}\Omega}{3} = 1.7\text{k}\Omega$$

Parallel Circuit Characteristics

Current Characteristics

$$I_T = I_1 + I_2 \ldots + I_n$$

Where, I_n = the current through the highest-numbered branch in the circuit

Voltage Characteristics

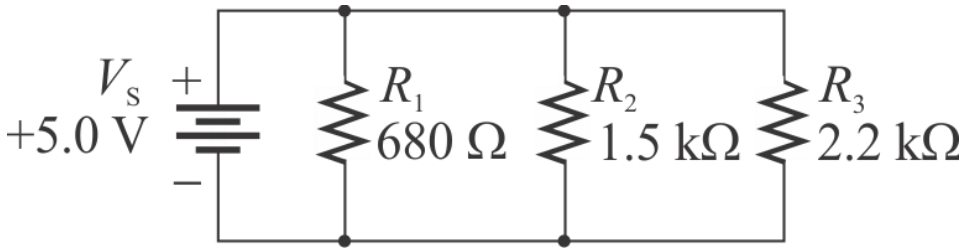
- Voltage across each component is equal
- Current through each branch is determined by the source voltage and the resistance of the branch.

$$V_S = V_1 = V_2 = \ldots = V_n$$

Power Characteristics

- Total Power – sum of the power dissipation values for the individual components

$$P_S = P_1 + P_2 \ldots + P_n$$

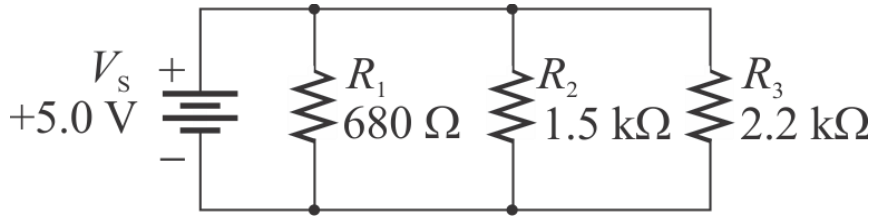


Tabulating current, resistance, voltage and power is a useful way to summarize parameters in a parallel circuit.

Continuing with the previous example, complete the parameters listed in the Table.

I_1= 7.4 mA	R_1= 0.68 kΩ	V_1=5.0 V	P_1= 36.8 mW
I_2= 3.3 mA	R_2= 1.50 kΩ	V_2= 5.0 V	P_2=16.7 mW
I_3= 2.3 mA	R_3= 2.20 kΩ	V_3=5.0 V	P_3=11.4 mW
I_T= 13.0 mA	R_T = 386 Ω	V_S= 5.0 V	P_T= 64.8 mW

Kirchhoff's Current Law (KCL)



Kirchhoff's current law (KCL) is generally stated as:

The sum of the currents entering a node is equal to the sum of the currents leaving the node.

Notice in the previous example that the current from the source is equal to the sum of the branch currents.

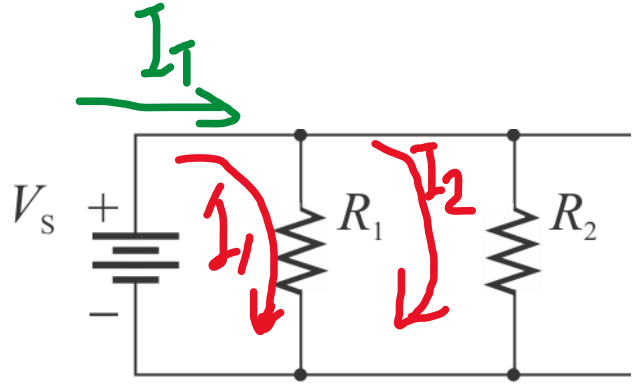
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I_T= 13.0 mA	R_T = 386 Ω	V_S = 5.0 V	P_T = 64.8 mW

Current Divider

When current enters a node (junction) it divides inversely proportional to each branch resistance.

The most widely used formula for the current divider is the two-resistor equation. For resistors R_1 and R_2,

$$I_1 = \left(\frac{R_2}{R_1 + R_2}\right) I_T \quad \text{and} \quad I_2 = \left(\frac{R_1}{R_1 + R_2}\right) I_T$$



Notice the subscripts. The resistor in the numerator is not the same as the one for which current is found.

Example

Assume that R_1 is a 2.2 kΩ resistor that is in parallel with R_2, which is 4.7 kΩ. If the total current into the resistors is 8.0 mA, what is the current in each resistor?

Solution

$$I_1 = \left(\frac{R_2}{R_1 + R_2}\right) I_T = \left(\frac{4.7\ \text{k}\Omega}{6.9\ \text{k}\Omega}\right) 8.0\ \text{mA} = \quad 5.45\ \text{mA}$$

$$I_2 = \left(\frac{R_1}{R_1 + R_2}\right) I_T = \left(\frac{2.2\ \text{k}\Omega}{6.9\ \text{k}\Omega}\right) 8.0\ \text{mA} = \quad 2.55\ \text{mA}$$

Notice that the *larger* resistor has the *smaller* current.

Recap

Series	In an electric circuit, a relationship of components in which the components are connected such that they provide a single path between two points.
Kirchhoff's voltage law	A law stating that the algebraic sum of the voltages around any closed path is zero.
Voltage divider	A circuit consisting of series resistors across which one or more output voltages are taken.

Recap

Parallel	The relationship in electric circuits in which two or more current paths are connected between two separate points (nodes).
Branch	One current path in a parallel circuit.
Kirchhoff's current law	A law stating the total current into a node equals the total current out of the node.
Node	A point or junction in a circuit at which two or more components are connected.
Current divider	A parallel combination of resistances in which the input current divides in inverse proportion to the branch resistances.